Thesis checklist

Content, clarity, and motivation issues only. Citation checking is deferred. Source line numbers may shift.

3.2.2 Bregman and beta-divergences

- □ **Repair the beta-divergence domain.** Reconcile the real power t^(1+beta) and regularity assumptions for arbitrary real beta > 0, including the separate quadratic case. [chapter 175-183]
- □ **Justify the symmetry characterization.** Establish that the stated assumptions justify the functional symmetry characterization, including the Legendre-type condition, or give a separate proof. [chapter 159-163]
- □ **Correct the scalar test functions.** Use the scalar t after defining p_t in the affine-invariance argument and justify the scalar conclusion in the symmetry proof. [chapter 170-172]

3.5.2 Proposition 9: proof

- □ **Introduce notation before it is used.** Define transformed coordinates and endpoints, m(t), q_0, sigma_N, and the maximizing-index sequence. Keep N and subsequence indices consistent. [chapter 620-649]
- □ **Justify the boundary row-sum bound.** The upper-bound argument permits an endpoint limit but refers to Proposition 8, whose statement assumes an interior limit. Explain the boundary case separately. [chapter 627-645; 575-580]
- □ **Correct the Rayleigh-bound notation.** Use b_N consistently and identify the correct index set. In the pair-count explanation, j is outside I_N; up to k indices can occur on each side. [chapter 666-675]
- □ **Correct the sign in the E_N bound.** Replace u-v by the interval length v-u in the lower bound for sigma_N times the number of indices, and check the limit. [chapter 677]
- □ **Connect the final index to the stated maximizer.** Identify the final i_N with the maximizing index used at the start and with k_N in the proposition, and make the limit argument explicit. [chapter 616-621, 681]
- □ **Resolve the empty appendix heading.** The proof now appears in the main chapter, leaving its appendix subsection empty. Resolve that heading and any relevant references. [chapter 620-682; appendix 495]